

# Towards a Cubic Size Bound for POSIX Lexing with Strong Simplification: An Incrementally Compiled Ledger of Checked Progress

Table 5 (Claude) agent session, with the Codex (GPT-5.5) supervisor agent  
repository `posix-codex`, branch `codex/backref-values`, theory `Posix` (Isabelle2025-2)

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## Abstract

This document is an incrementally updated ledger of the campaign to prove a cubic size bound for the Antimirov-factored strong-simplification route of the POSIX lexer formalization. Every statement labelled *checked* names an Isabelle lemma that builds green in the `Posix` session; every counterexample labelled *checked* is likewise an Isabelle fact. Mirror-validated statements (random testing against a clause-verified Python model) are clearly marked as such and are *not* proofs. The document is recompiled after each increment of progress.

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## 1 The target

For the erased regular-expression datatype `rrexpr` (constructors `RZERO`, `RONE`, `RCHAR`, `RSEQ`, `RALTS`, `RSTAR`, `RNTIMES`, plus backreference constructors excluded from the fragment), the live target is the *set-ledger cubic gate* (`MAINLINE.md`):

$$\text{rsize\_set}(\text{row\_dlformss}(\text{rpder\_strong\_rows\_raw } c(\text{afactored1 } r \ s))) \leq 2(\text{rsizer} + 3)^3.$$

In words: take the rows produced by one strong derivative step from the factored front of  $r$  after input  $s$ , open every row into its linear forms, deduplicate *across the whole union*, and sum the sizes of the distinct opened rows; this total must be cubic in the size of the original regex. POSIX value preservation is handled by the separately checked deferred-memo reconstruction theorems and is not re-proved here.

## 2 Checked refutation: the duplicated list account is exponential

The first reduction attempt measured the *duplicated list* cost `afactored1_strong_dlform_list_cost` (sum of opened-list sizes, duplicates counted). This is provably hopeless.

**Theorem 2.1** (checked: `afactored1_strong_dlform_list_cost_cubic_false`). *There is a family  $r_j$  with  $\text{rsizer}_j = \Theta(j)$  whose one-step duplicated list cost grows as  $2^j$ ; in particular the cost is not bounded by  $2(\text{rsizer} + 3)^3$ , nor by any polynomial.*

The witness is the *RONE-pair tower*: with  $m' = \text{RSEQ}(\text{RALTS}[\text{RONE}, a])(\text{RALTS}[\text{RONE}, b])$  and  $x_j = \text{RSEQ}(\text{RALTS}[\text{RONE}, m'])x_{j-1}$ , every nesting level doubles the number of opened rows because the two RONE-paths of  $m'$  both reopen the tail; the tower is a literal fixpoint of the strong simplifier, so simplification does not help. All fully collapsing RONE-paths produce *the same* row, which is why the deduplicated (set-level) account below survives.

## 3 Checked set-level theory

**Theorem 3.1** (checked: `card_row_dlforms_le_rsize`). *For every row  $q$ :  $|\text{row\_dlforms } q| \leq \text{rsize } q$ . (Distinct opened rows of a single row are at most linear, unconditionally.)*

The proof is a set-difference triangle along the reassociation chain; the core checked inequality is `card_row_dlforms_rsimp4_diff_le`:  $|\text{row\_dlforms}(\sigma_4(p, k)) \setminus \text{row\_dlforms } k| \leq \text{rsize } p$ , where  $\sigma_4$  is the sequence-atom smart constructor.

**Theorem 3.2** (checked: `rsize_set_row_dlforms_le_rsize_sq`).  $\text{rsize\_set}(\text{row\_dlforms } q) \leq (\text{rsize } q)^2$  for every row  $q$ .

**Theorem 3.3** (checked: `card_row_dlformss_le_rsizes` and `card_row_dlformss_rpder_strong_rows_raw_le_generated`). *The deduplicated union over a row list has at most  $\text{rsizes}(\text{rows})$  members; instantiated to the actual one-step output, the union card is at most the generated total size (one degree).*

## 4 Checked gate reduction: from three summands to two numbers

The supervisor's pair-budget gate splits the union ledger into a non-sequence part, a front-times-tail-weights part, and an active-suffix pair-budget part. Both abstract premises of the gate were discharged (`actual_union_seq_head_size_linear` gives head bound  $H = \text{Suc}(2 \text{rsizer})$  on the legacy fragment; `rseq_tails_row_dlformss_rpder_strong_weight_le_generated` gives the key weight  $M$ ), yielding the instantiated gate `actual_union_pair_budget_gate_instance`. Then:

**Theorem 4.1** (checked: `pair_budget_afactored1_strong_dlform_universe_zero`). *The active-suffix pair budget of the dlform universe is identically zero: opened rows always have non-alternation heads (`row_dlforms_seq_member_head_nonalt`), so no member of the universe has the keyed shape  $\text{RSEQ}(\text{RALTS rows}) k$ . The third gate summand vanishes (`actual_union_gate_two_summands`).*

**Theorem 4.2** (checked: exact decomposition). *With `rsizet.set.split.rnonseq.rseq`, `rsizet.set.rseq.members.le.h` and `sum.rseq.member.tails.regroup`, the whole gate rests on exactly two numbers for the actual opened union  $U$ :*

$$\text{rsizet.set}U \leq \underbrace{\text{nonseq part}}_{\text{cubic, checked}} + |\text{rseq.members}U| \cdot \text{Suc } H + \sum_t |\text{rseq.tail.rows}U t| \cdot \text{rsizet}.$$

*The non-sequence part is cubic by `rsizet.set.rnonseq.members....cubic` via the front-terms carrier.*

## 5 Checked static universe results

**Theorem 5.1** (checked: `rsizes.afactored1.le.rsizet.set.apder.rows`). *For every input  $s$ : `rsizes(afactored1 r s) ≤ rsizet.set(apder.rows r)`. The multi-step front total is bounded by a static, input-independent quantity (rows are distinct and live in the static row universe).*

**Theorem 5.2** (checked: `apder.rows.member.size.quadratic`). *On the `apder.nf` fragment every member of the static row universe has size at most  $\text{Suc}((\text{rsizer} + 2)^2)$ . Hence every dynamic front row is at most quadratic — the row-size half of the front-quadratic conjecture.*

**Counterexample 5.3** (mirror-verified, tight). *The static ledger `rsizet.set(apder.rows r)` is cubic-tight: on the star-seq towers  $x_{i+1} = (x_i b_i)^*$  the ratio `rsizet.set/(rsizet)`<sup>3</sup> converges to  $\approx 0.040$ . The checked cubic bound `rsizet.set.apder.rows.expanded.cubic.size.bound` cannot be improved to quadratic. Moreover the reachable row universe equals the static one on this family, so no set-shaped (history-independent) bound can beat cubic; the missing degree is irreducibly about rows coexisting in one front (liveness).*

## 6 The row-count law (the “D law”)

The remaining static improvement is row-count linearity, classical Antimirov shape. Write `accrk` for `apder.term.frontier.acc`, `Fk` for `rfrontier k`.

**Definition 6.1** (checked: `apder.zw2`).  $\text{zw}_2(\text{RCHAR}) = 1$ ; sums over `RALTS/RSEQ`;  $\text{zw}_2(\text{RSTAR } x) = 1 + \text{zw}_2(x)$ ;  $\text{zw}_2(\text{RNTIMES } r \ n) = n(1 + \text{zw}_2 r)$ ; 0 otherwise. Checked:  $\text{zw}_2 \leq \text{rsizet}$  on the `rntimes.free` fragment (`apder.zw2.rntimes.free.le.rsizet`).

**Conjecture 6.2** (the D law, `rntimes.free` form; mirror: > 295,000 deep samples, zero violations). *On the legacy, `apder.nf`, `rntimes.free` fragment:*

$$|\text{accrk} \setminus \text{Fk}| \leq \text{zw}_2 r.$$

### 6.1 Checked falsifications along the way

**Counterexample 6.3** (checked and mirror; weight too small). *The plain letter-count (`apder.awidth`) version is false: zero-width stars produce frontier rows without letters. The first corrected weight (`apder.zwidth`, `star = max 1`) is also false at depth:  $r = b(\text{ALTS}[(a^*)^*, (b^*)^*, b])$ ,  $k = \text{RONE}$  gives  $5 > 4$  — each nested zero-consuming star layer needs its own slot.*

**Counterexample 6.4** (checked: `apder.zw2.D.law.rntimes.zero.alt.false`). *The unrestricted  $\text{zw}_2$  law is false with zero counted repetitions:  $r = c(\text{ALTS}[\text{RNTIMES } a \ 0, \text{RNTIMES } b \ 0])$  has two imported frontier atoms but  $\text{zw}_2 r = 1$ . Hence the `rntimes.free` restriction in Conjecture 6.2.*

Nine inductive strengthenings (subset/intersect discounts, membership companions, potential and reserve forms, an unduplicated list-length form, and the first joint invariant  $J^*$ ) were falsified by explicit counterexamples; the full list with witnesses is maintained in `MATHPROBLEM_ROWCOUNT.md`. A hard methodological lesson is recorded there: two false statements each passed  $\geq 95,000$  *shallow* random samples; all sampling now requires depth  $\geq 5$  plus directed nested-star/zero-width families.

## 6.2 Constructor bridges (checked, supervisor)

The current attack decomposes the D law by head constructor. Checked bridges so far include the RSEQ frontier singleton shifts, the  $\sigma_4$  non-RONE preservation and frontier-shape lemmas, the normal-nonalt frontier/carry-bucket shifts, the right-nonalt SEQ D bridge, and its RCHAR-left specialization (`card_apder_term_frontier_acc_RSEQ_RCHAR_diff_le_if_right_nonalt_diff`): an ordinary D bound for an `apder_nf` non-alternation right child yields the exact parent SEQ bound when the left child is a character.

## 6.3 Payoff when landed

$|\text{apder\_rows } r| \leq \text{zw}_2 r + 2$ , hence with the checked quadratic member bound:  $\text{rsizes}(\text{afactored1 } r \text{ } s) \leq (\text{zw}_2 r + 2) \cdot \text{Suc}((\text{rsizer} + 2)^2)$  — a *cubic static front bound by assembly* on the fragment, replacing the tight-ledger detour, and the row-count half of the dynamic front-quadratic conjecture.

## 7 The liveness frontier

All single-step syntactic quantities are now known cubic-tight (static ledger, per-member size, per-row norm ledgers — each with explicit witness families). The remaining degree for the full gate is the *liveness slice*: simultaneously live rows are a quadratic slice of the cubic static universe. Mirror-validated with zero violations across all families is the cap-law invariant

$$\text{rsizes}(\text{afactored\_step } c \text{ } rows) \leq \max(\text{rsizes } rows, C(\text{rsizer})^2),$$

factored into three single-step obligations (jump bound, monotone self-preservation, jump saturation) — the first formulation in which every obligation quantifies over one front and one step. Formalization has not started.

## 8 Progress log

| When       | Increment                                                                                                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-06-12 | Checked: atom multiplicity, double counting, per-front-row payment interface. Checked refutation: RONE-pair tower kills the duplicated list-cost target (exponential).                                                               |
| 2026-06-12 | Checked: set-level per-row theory (card linear, ledger quadratic); union card account; weight and head bounds; pair-budget gate instantiated, then its budget proven zero; gate cut to two summands; exact two-number decomposition. |
| 2026-06-12 | Checked: front staticized ( $F \leq A$ for all inputs); static maxrow quadratic. Mirror: static ledger cubic-tight; reachable = static; cap-law invariant zero violations; liveness reduced to three single-step obligations.        |

| When          | Increment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-06-12/13 | Row-count law campaign: weights <code>awidth/zwidth</code> falsified (checked CEs), <code>zw2</code> landed; nine strengthenings falsified; deep-sampling standard instituted; <code>rntimes_free</code> target fixed. Supervisor: constructor-bridge decomposition under way (RCHAR-left SEQ bridge checked).                                                                                                                                                                                                                                                                                                               |
| 2026-06-13    | This document created; compiled with <code>pdflatex</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2026-06-13    | Supervisor (overnight-morning): <code>apder_zw2</code> landed with <code>rntimes_free</code> payoff; checked CE <code>apder_zw2.D_law_rntimes_zero_alt_false</code> restricts the D law to the <code>rntimes_free</code> fragment; constructor-bridge decomposition: RSTAR and RALTS branches of the D induction fully checked ( <code>card_apder_term_frontier_acc.RSTAR.diff_le, ..._RALTS.diff_le_if_children</code> ), RCHAR-left SEQ bridge checked. The Suc in <code>zw2(RSTAR)</code> pays exactly the singleton frontier of the composed continuation.                                                               |
| 2026-06-13    | <i>J*</i> revived under <code>zw2</code> : the joint invariant (left-acc minus own composite frontier + imported-uncovered + sibling count $\leq zw_2r_1 + zw_2r_2$ ) passes the post-correction sampling standard — 119,207 deep random samples (66,719 non-trivial) plus a 343-case directed grid, zero violations. It implies the SEQ branch of the D law including the hard right-RALTS degenerate-continuation case. Status: mirror-validated conjecture, recommended joint-induction skeleton.                                                                                                                         |
| 2026-06-13    | Checked: <code>card_apder_term_frontier_acc.RSEQ.diff_le_if_jstar</code> — the <i>J*</i> joint invariant implies the general SEQ step of the D law by a three-block covering of the union difference (pure set algebra). Any future proof of <i>J*</i> - <code>zw2</code> closes the SEQ branch outright; together with the checked RSTAR/RALTS/base branches, the D induction then assembles.                                                                                                                                                                                                                               |
| 2026-06-13    | New named invariant E0- <code>zw2</code> (merged-frontier law), mirror-validated at full standard (171,161 deep samples + directed grid, both the $k = \text{RONE}$ and general- $k$ forms): $ (F(\sigma_4(q, k)) \cup \text{acc } q \ k) \setminus Fk  \leq 1 + zw_2 q$ . Checked: <code>jstar_RCHAR_if_E0</code> — E0 closes the RCHAR base case of the <i>J*</i> induction by disjoint-union set algebra. D-induction map: base/RCHAR/RALTS/RSTAR checked, SEQ checked conditional on <i>J*</i> ; <i>J*</i> base checked conditional on E0. The frontier is now E0 (single-variable!) plus the <i>J*</i> inductive steps. |